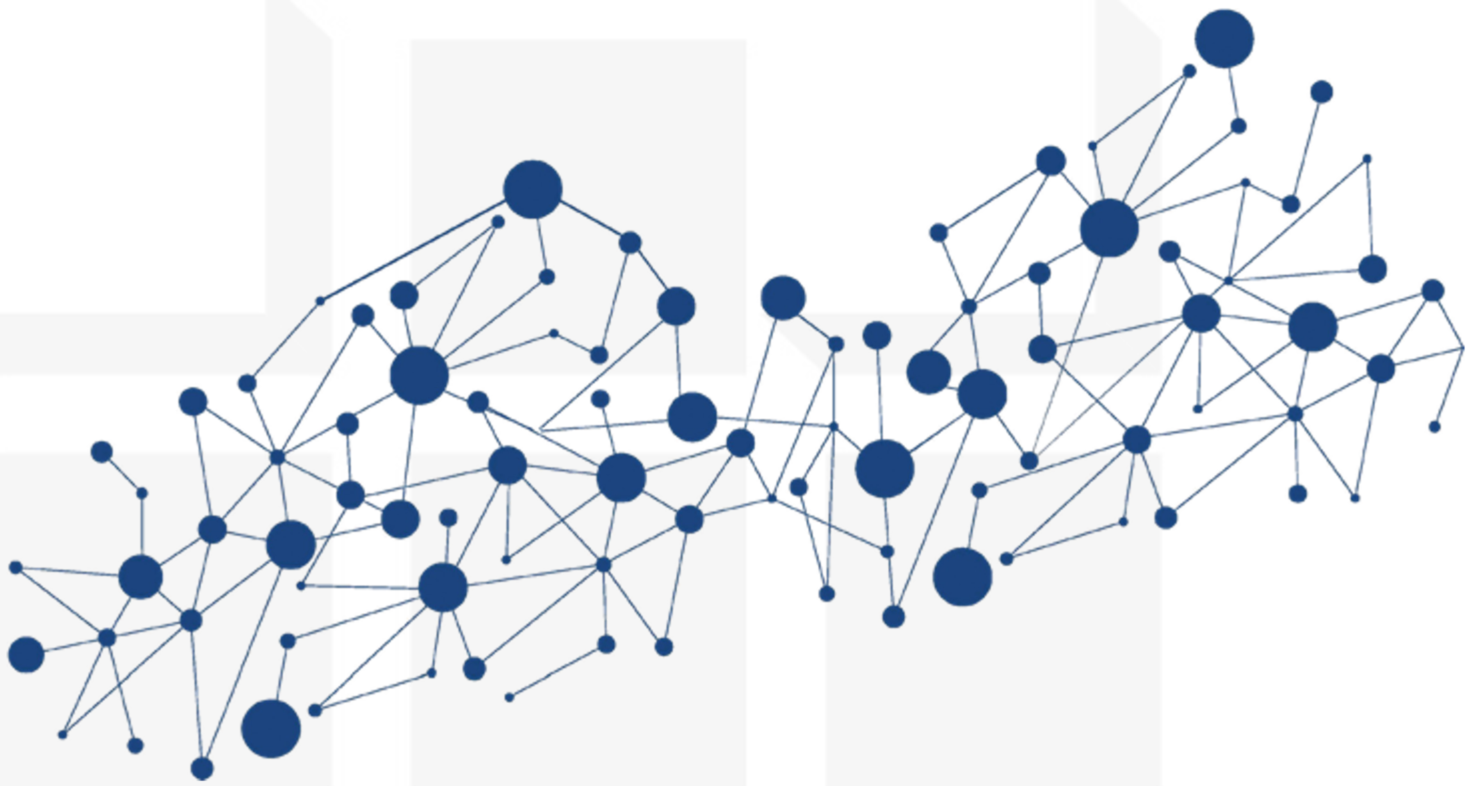
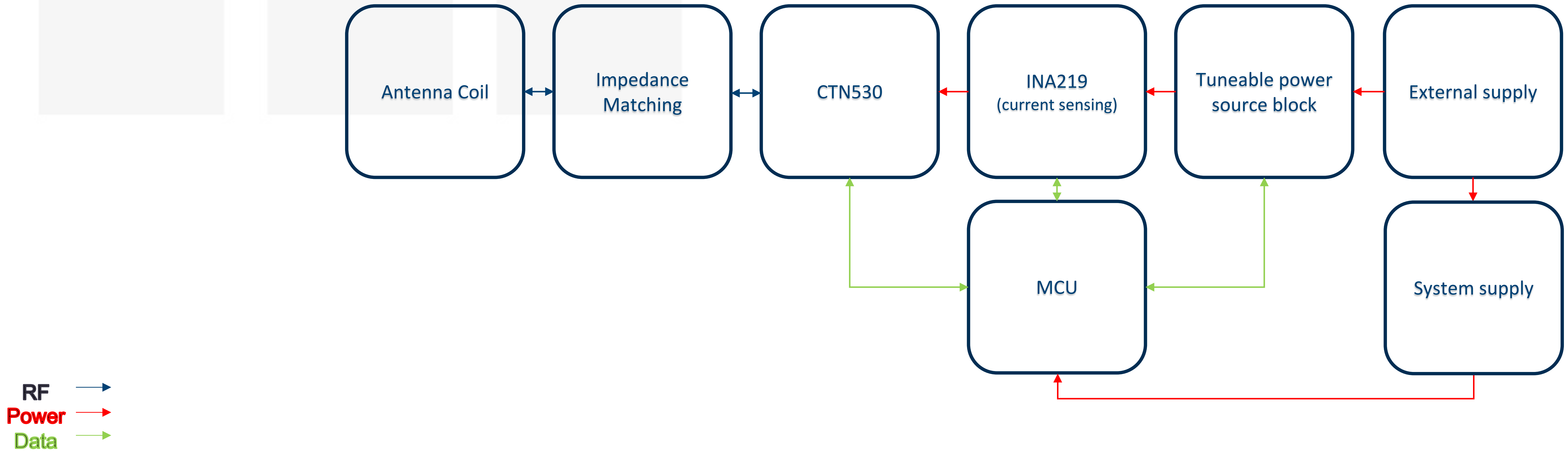


Poller

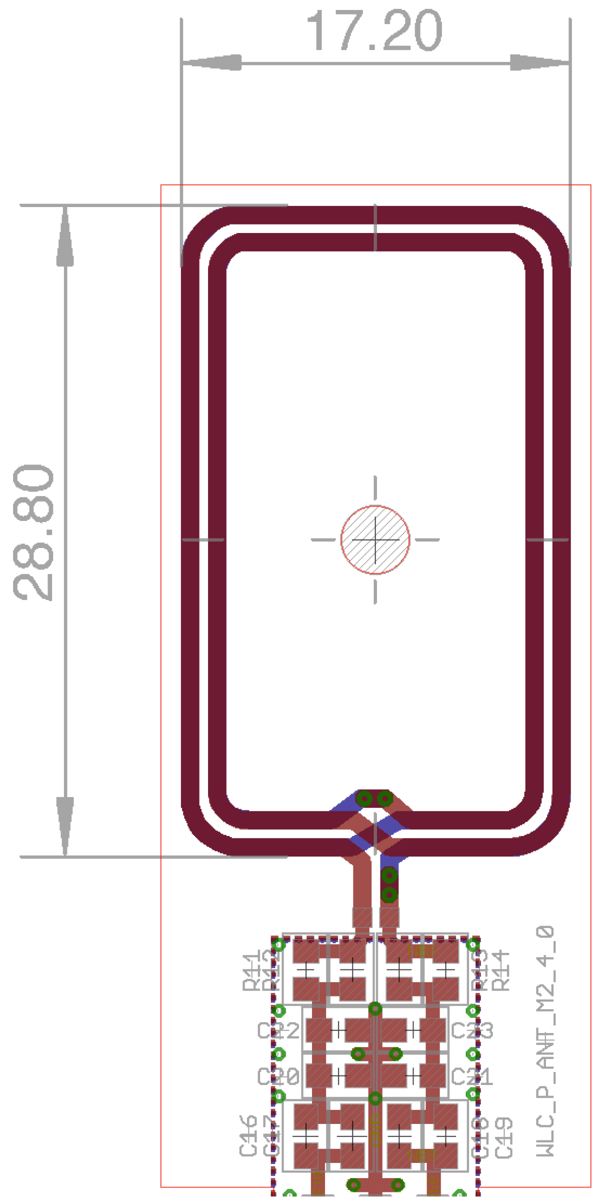
Must have components

- The simplified block diagram of must have components for enabling the WLC system designed by CISC for Sibel is shown bellow.
- Any other components (like EEPROM, Leds, Sensors, etc.) are considered as optional.



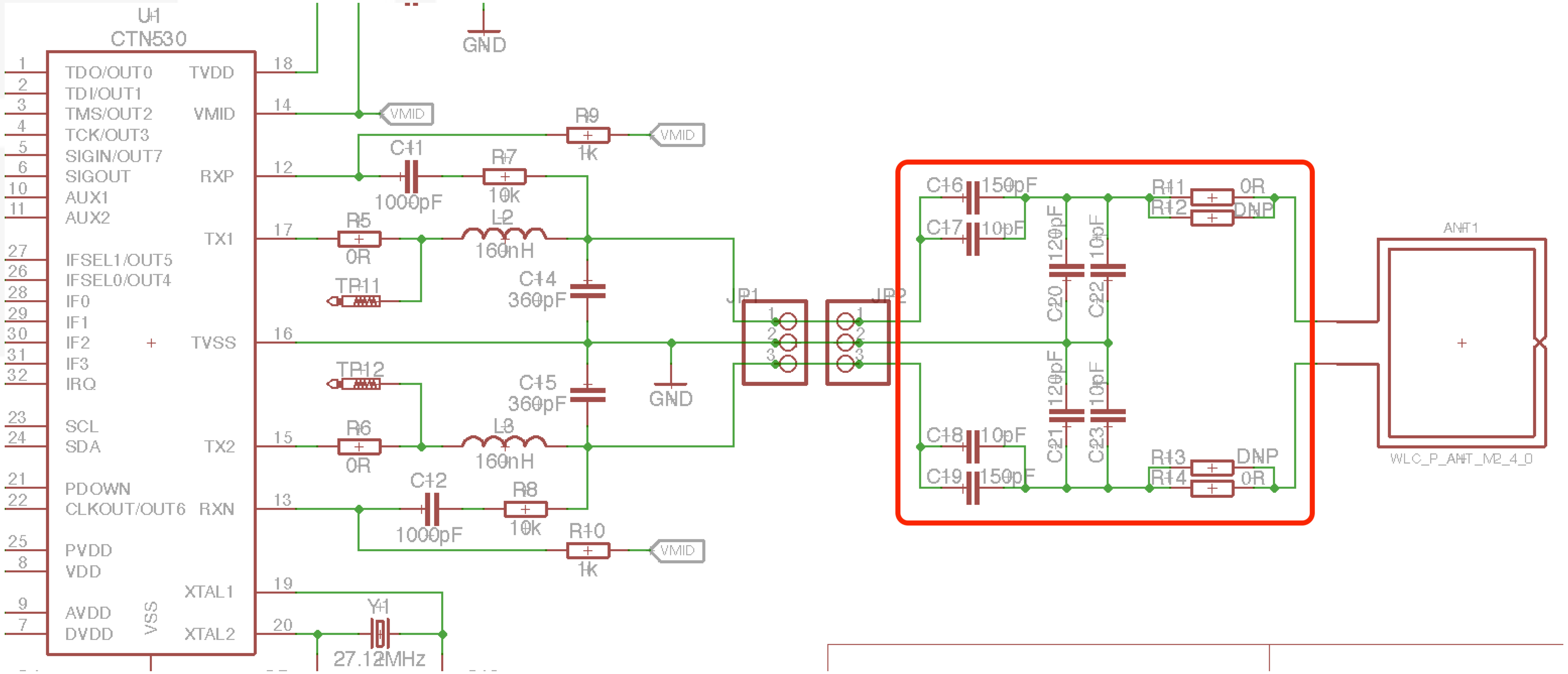
Antenna

- Customized antenna has been designed:



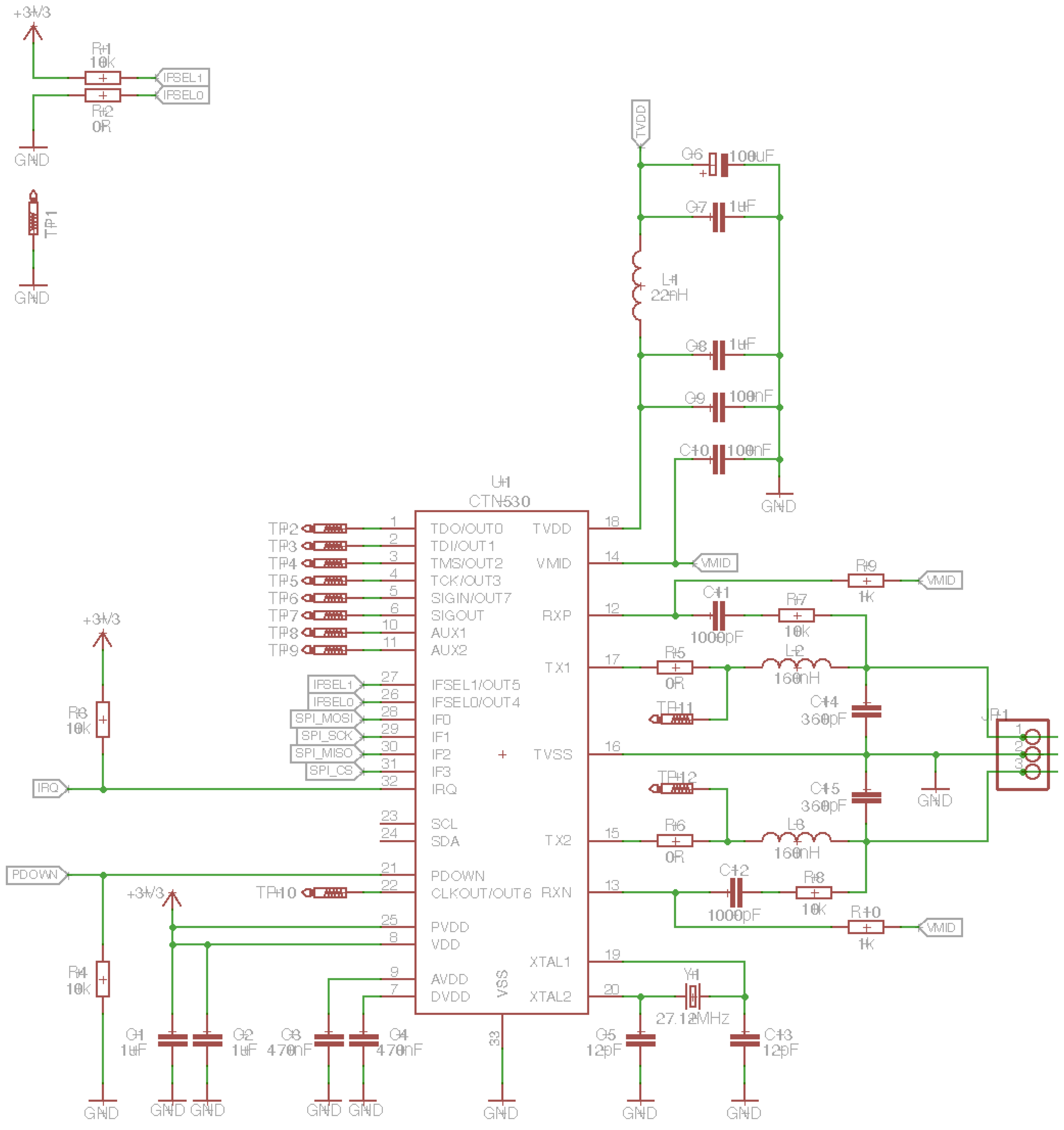
Impedance Matching

- The designed matching structure together with the value of the components:



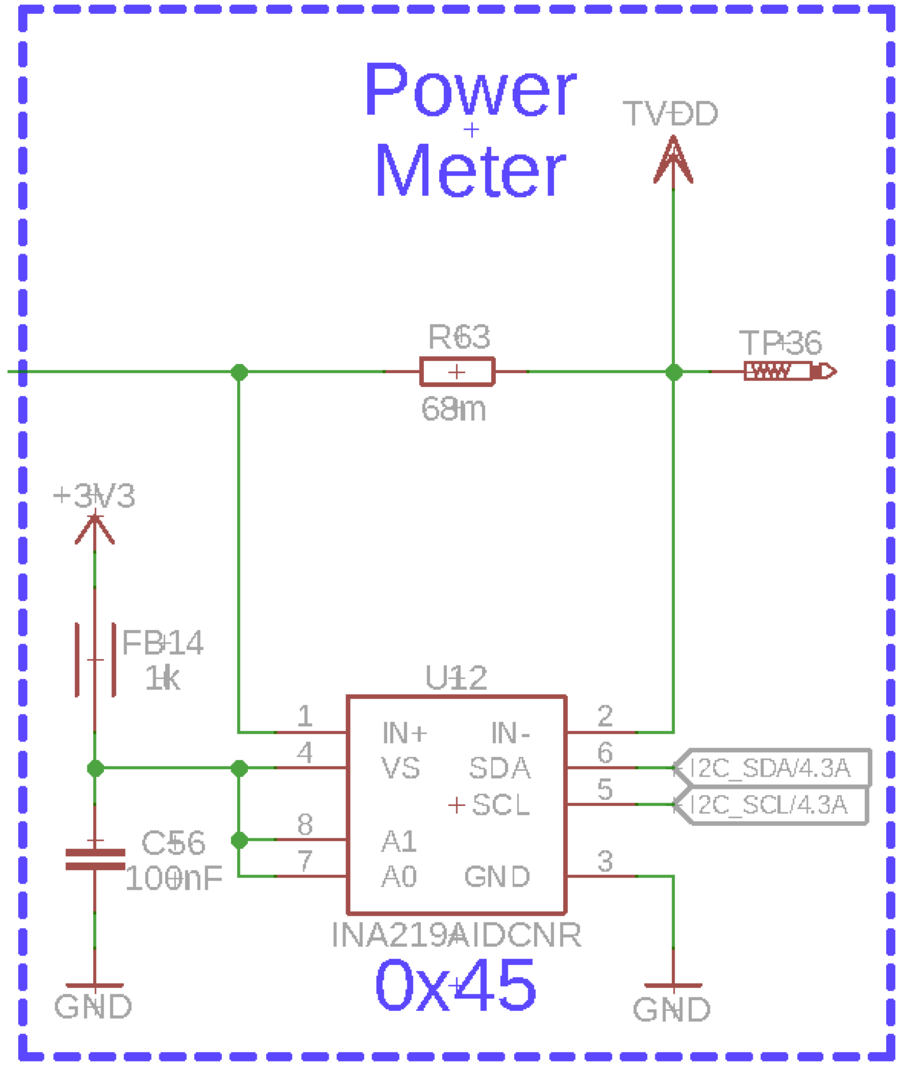
CTN530

- The design of the CTN530, which is the NFC Wireless Charging Transmitter Front-End:
  - ❖ IFSEL1 and IFSEL0 enables the SPI interface
  - ❖ SPI_MOSI, SPI_SCK, SPI_MISO, SPI_CS is the communication SPI interface, up to 10 Mbit/s.
  - ❖ IRQ is a must-have pin connected to the MCU GPIO Input pin.
  - ❖ PDOWN is a must-have pin connected to the MCU GPIO Output pin.
  - ❖ PVDD and VDD are the digital core supply used by the system supply (supported 2.5 – 5.5V).
  - ❖ TVDD is the Transmmitter voltage supply generated by the “Tunable power source block”.



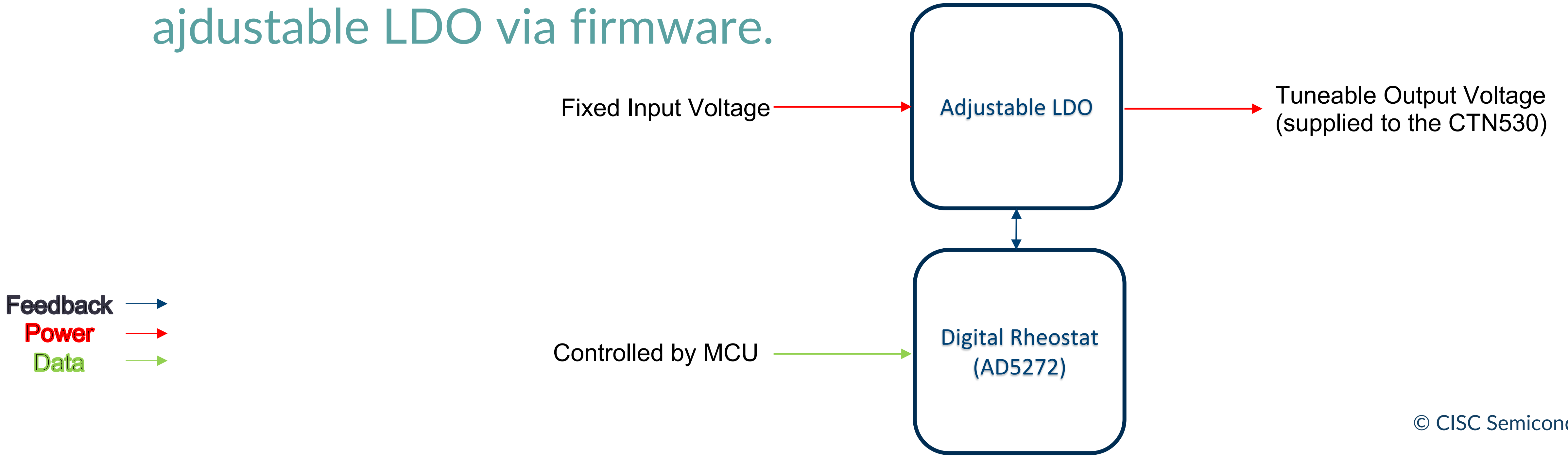
INA219 (current sensing)

- INA219 is a 12-bit, i2c output current/voltage/power monitor.
- It's main usage is to measure the current consumption of the CTN530 transmitter during the Wireless Power Transfer (WPT) used for the background Foreign Object Detection.
- Theoretically a different current monitor IC can be used, but currently only INA219 is supported.
- Current sense resistor is connected in a way to be able to measure the current drawn by the TVDD pin of the CTN530.
- Example of the INA219 connection ------------------->



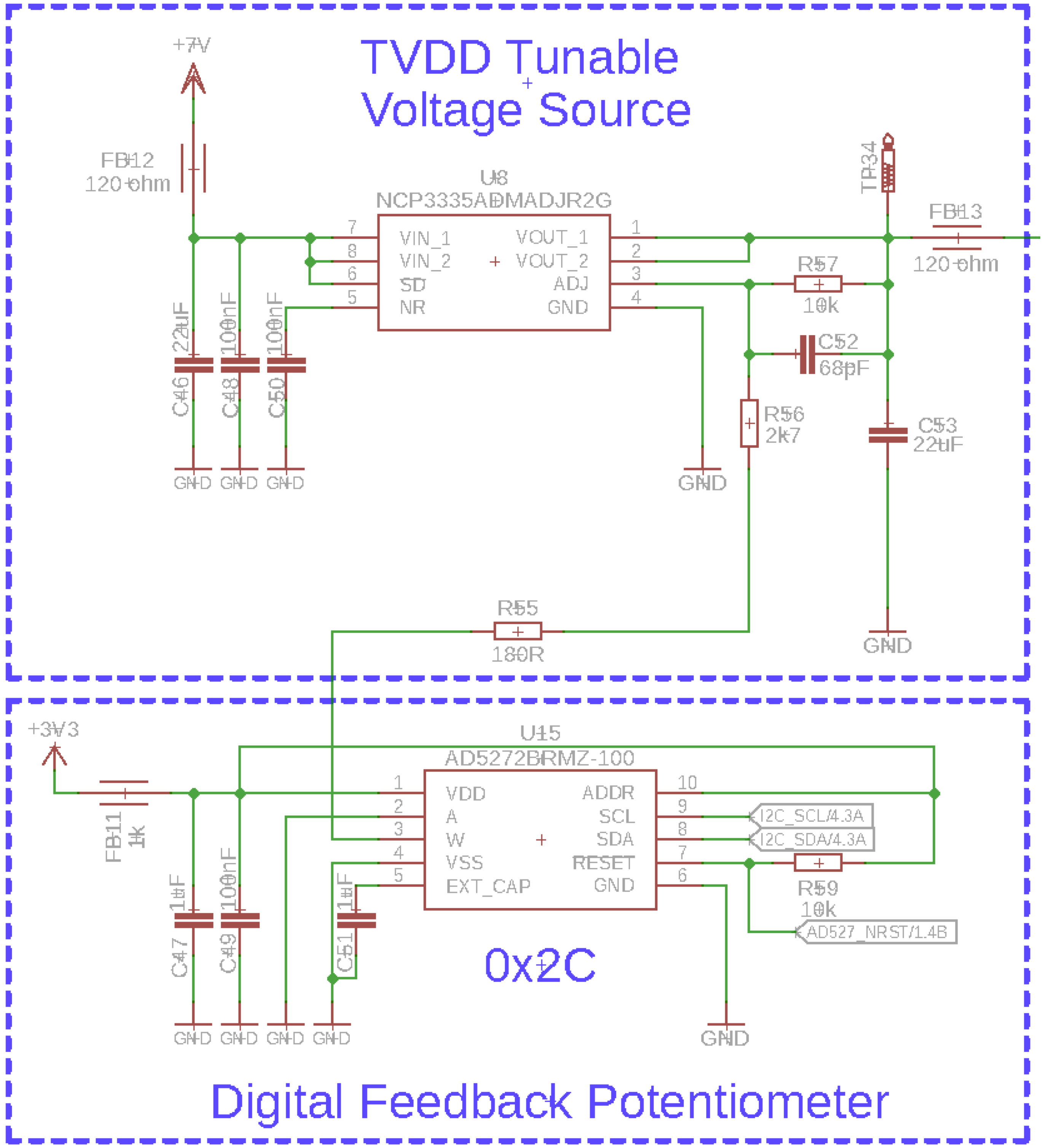
Tuneable Power Source Block (1/2)

- The WLC system supports multiple power steps used for the charging and communication. In order to be able to provide these power steps, the Poller desing contains a Tuneable Power Source Block. The power source block is designed by a multiple components and controled by the Poller firmware.
- The main part of this block is a AD5272 - Digital Rheostat controled via I2C.
- The simplifed schematic of the Tuneable Power Source Block:
  - ❖ The adjustable LDO sets the Output voltage based on the Feedback Resistor, which is realised by AD5272, digital Rheostat. Thus, the MCU can control and vary the output voltage of the ajdustable LDO via firmware.



Tuneable Power Source Block (2/2)

- As adjustable LDO, the NCP3335ADMADJR2G is used. The output voltage is calculated as Vout = 1.25 * (1 + R1/R2), where R2 is the AD5272 (100kΩ) in series with 180Ω + 2k7Ω, and R1 = 10kΩ.
- Thus, the output voltage is in range 1.37 - 5.6V (The CTN530 supports 2.5 – 5.5V).
- Theoretically a different Digital Rheostat IC can be used, but currently only AD5272 is supported.
- Any other adjustable LDO can be used, as long as the Vout uses the same formula. Otherwise, firmware macro needs to be modified.

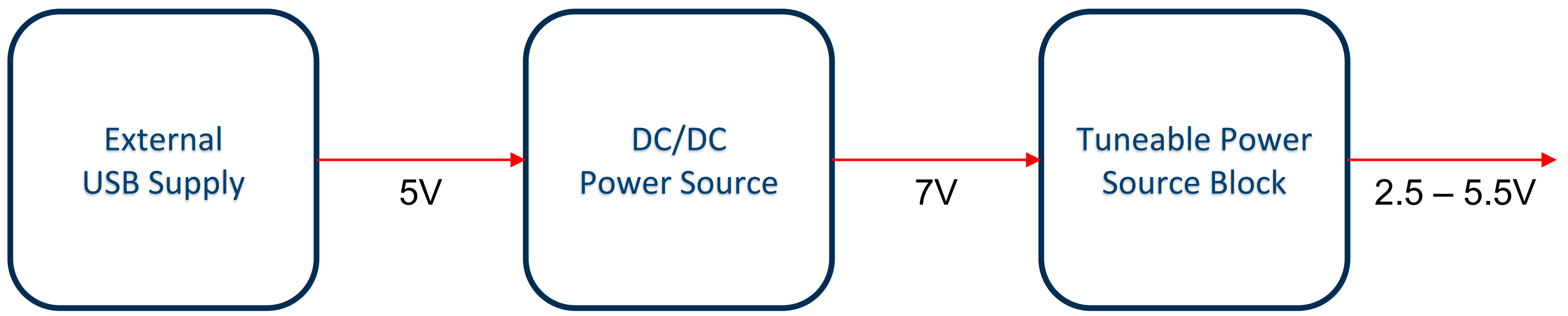


External Supply 1/2

- Since the Poller is WLC transmitter and the main function is to charge the Listener (WLC receiver), an external power supply is required - theoretically any source capable of providing sufficient current (up to 350 mA) for a long period of time, can be used. Typically a USB is used.
- In order to use full range of power of the designed system, the External Supply must be able to create sufficient voltage for the Tuneable Power Source Block, which is used to supply the CTN530 transmitter. Here, the full range is 2.5 – 5.5V. Therefore, an External Supply should be able to provide at least 5.5V + Voltage drop of the the Tuneable Power Source Block.
- Typically, 7V is supply with sufficient margin.

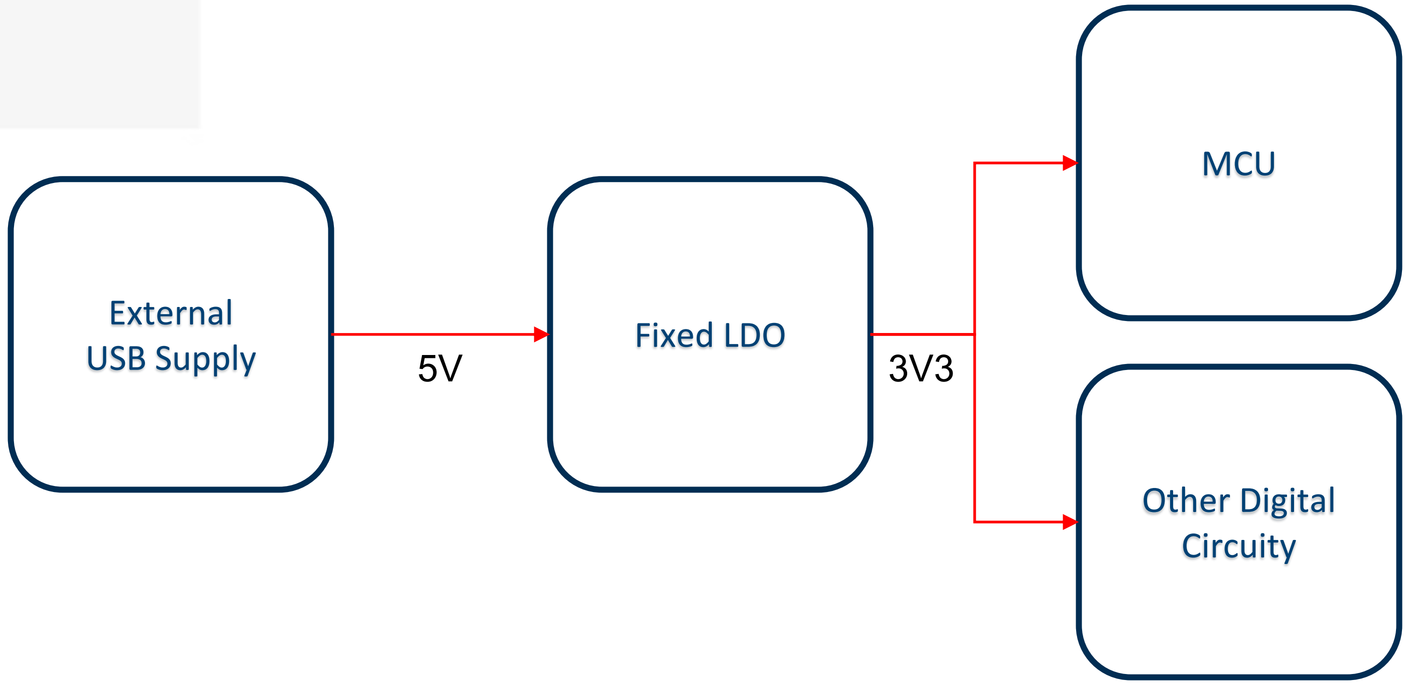
External Supply 2/2

- The CISC WDK BaseStation uses a USB as a main power supply and then DC/DC booster to create a 7V, as a supply for the Tuneable Power Source Block.
- Any DC/DC (or any other solution) can be used.
- Simplified block example of the External Supply together with the Tuneable power Source block used in the WDK BaseStation:



System Supply

- This is theoretically an optional part, but if the design should not use directly the External Power Supply (such as the USB) to supply the system, an additional DC/DC (or LDO) is required.
- Any type of a DC/DC (or LDO) can be used.
- Simplified block example of the System Supply used in the CISC WDK BaseStation:



MCU

- MCU must run a WLC Host firmware.
- Theoretically any type of a MCU can be used, since the WLC Host Library is portable.
- For Sibel, WLC Host Library has been ported for the Nordic nRF52840 MCU, running the Zephyr OS.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